BINGNAN CHEN

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About Me

I am a Ph.D. student in Computer Science and Engineering at the Hong Kong University of Science and Technology (HKUST), where my research focuses on database optimization algorithms, query processing, and query optimization under the supervision of Prof. Ke Yi. I previously obtained my MPhil degree from HKUST and my bachelor's degree from the University of Science and Technology of China (USTC).

Education

- **The Hong Kong University of Science and Technology** *Sept., 2023*
  Computer Science, Doctor of Philosophy (PhD) Hong Kong, China
- **The Hong Kong University of Science and Technology** *Sept., 2021*
  Computer Science, Master of Philosophy (MPhil) Hong Kong, China
- **University Of Science and Technology of China** *Sept., 2017*
  Computer Science, Bachelor Hefei, China

Research

- **Outer Join optimization.** *Nov. 2025 - Oct. 2026*
  Tools: C++
  - Develop and implement optimization algorithms for **Outer Join and Join-Aggregate Queries**, with provable theoretical guarantees and a focus on efficient query processing. Integrate and evaluate the proposed optimizations within DuckDB's core query optimizer, ensuring compatibility with its existing optimization framework.

- **Engineering Yannakakis$^+$ in DuckDB: From Theory to Practice** *Mar. 2025 - Nov. 2025*
  Tools: C++
  - Develop and integrat **Yannakakis$^+$** into the core query optimizer of DuckDB, a push-based, vectorized, and pipeline-interpreted OLAP database system. This implementation replaces the native dynamic programming join optimizer with a **greedy GYO-based plan enumerator** tailored for acyclic queries, employs **Bloom filters** to replace expensive semi-joins, and leverages DuckDBs advanced statistics and cost models for **hybrid CBO/RBO** plan selection. The system demonstrates that output-sensitive join algorithms can be made both asymptotically optimal and practically efficient in modern analytical engines.

- **Yannakakis$^+$: An optimization algorithm for SPJA queries** *Sept. 2023 - Sept. 2024*
  Tools: Python, Java & Scala
  - We proposes **Yannakakis$^+$**, a practical variant of the classic Yannakakis algorithm for evaluating acyclic conjunctive queries with aggregations. It preserves the theoretical guarantees of **O(N+M)** time on free-connex queries and **O(min(NM, F))** on general acyclic queries, while reducing constant factors by interleaving selective **aggregation-joins** with a reduced number of **semi-joins** and by optimizing the plan selection in a tailored search space. The approach yields pure relational DAG plans built from standard operators and is implemented as a plug-in for multiple SQL engines, showing substantial speedups across a wide range of queries and datasets.

- **Quorion: Interactive Web UI System for Query Rewriting** *Sept. 2024 - Mar. 2025*
  Tools: Python, Java & Scala
  - We present **Quorion**, a system that rewrites SPJA SQL queries into sequences of SQL statements that implement nextgeneration join/aggregate plans with theoretical guarantees, notably **Yannakakis$^+$** for acyclic queries, as well as GHD- and GYO-based reductions for broader cases. The system includes a cost- and rule-based optimizer, a **plan visualization** interface for selecting join trees, and a portable execution model that targets existing DBMSs (DuckDB, PostgreSQL, MySQL) via JDBC. A motivating example and a brief demonstration workflow suggest substantial performance gains on certain hard queries without modifying DBMS internals.

- **Flexway O-Sort: Oblivious sorting algorithm under SGX.** *Aug. 2022 - May. 2023*
  Tools: C++, SGX
  - **Flexway O-Sort** is an optimal oblivious (data-independent, memory access patterns do not reveal any information about the input data even to an untrusted observer) sorting algorithm designed for secure hardware enclaves. It achieves $O(N \log N)$ work and optimal page swaps through a novel **flexway butterfly network** and a **p-way Merge-Split** that runs efficiently and is **enclave-friendly**. The algorithm is **parallelizable** under secure settings, and significantly outperforms prior oblivious sorts.

EXPERIENCE

- **Alibaba Cloud [🌐]** *June 2026 - Present*
  Research Intern Hangzhou, China
  ◦ Implement and deploy the Yannakakis$^+$ algorithm for runtime optimization in query execution engine.
- **Alibaba Cloud [🌐]** *Aug. 2023 - Feb. 2024*
  Research Intern Hangzhou, China
  ◦ Design and implement the Yannakakis$^+$ algorithm, which is an improved version of Yannakakis algorithm with theoretical guarantees and better practical performance.
- **SenseTime [🌐]** *Jan. 2021 - Jul. 2021*
  Research Intern Beijing, China
  ◦ Design operators and compilation optimizations in deep learning frameworks.

PUBLICATIONS

C=CONFERENCE, J=JOURNAL, S=IN SUBMISSION, T=THESIS

[S.1] **Outer Join Optimization**. **First author**. *Submitted to ACM SIGMOD 2027 (Under Review, title withheld for double-blind review).*

[S.2] **Bingnan Chen**, Qichen Wang, Binyang Dai, Ke Yi, Feifei Li, and Liang Lin. Yannakakis$^+$: Practical Acyclic Query Evaluation with Theoretical Guarantees. Submitted to ACM Transactions on Database Systems (TODS 2026). Manuscript under revision.

[C.1] Qichen Wang*, **Bingnan Chen***, Binyang Dai, Ke Yi, Feifei Li, and Liang Lin. Yannakakis$^+$: Practical Acyclic Query Evaluation with Theoretical Guarantees. In Proceedings of the 2025 International Conference on Management of Data (Berlin, Germany) (SIGMOD 25) *(* equal contribution)* [pdf].

[C.2] **Bingnan Chen**, Binyang Dai, Qichen Wang, Ke Yi. Query running too slow? Rewrite it with Quorion!. In Proceedings of the 51st International Conference on Very Large Data Bases, Demo Track (London, United Kingdom), (VLDB 25) [pdf].

[C.3] Tianyao Gu, Yilei Wang, Afonso Tinoco, **Bingnan Chen**, Ke Yi, and Elaine Shi. Flexway O-Sort: Enclave-Friendly and Optimal Oblivious Sorting. In Proceedings of the 34th USENIX Security Symposium (Seattle, Washington, United States) (USENIX Security 2025) [pdf].

SKILLS

- **Programming Languages:** C++ & C, Python, Java, Scala, SQL, HTML & CSS & JSP, Verilog & System Verilog
- **Web Technologies:** Docker, Git, OOP, Django, Vue, Flask
- **Database Systems:** DuckDB, PostgreSQL, SparkSQL, MySQL
- **Data Science & Machine Learning:** PyTorch